

Harmonic Jets of Hypergeometric Sums: Inverse Parameter Lifting, Cyclotomic Holonomicity, and Lucas Descent

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Abstract

We develop a general calculus for finite sums in which a hypergeometric kernel is multiplied by a polynomial in ordinary or cyclotomic harmonic numbers. The first result is an inverse to the familiar operation of differentiating Pochhammer symbols: every harmonic polynomial of prescribed weight is a finite linear combination of mixed derivatives of explicit hypergeometric parameter products. An inclusion–exclusion product isolates a single letter $H_m^{(r)}$ without contamination by lower set partitions. It follows, after a finite residue-class split, that every finite proper hypergeometric sum with polynomial cyclotomic harmonic weight is P-recursive; the proof is constructive and reduces recurrence certification to ordinary creative telescoping.

The second result is arithmetic. We prove a character-resolved Lucas descent theorem: under a termwise digit factorisation and a transparent compatibility condition on the harmonic arguments, the weight- w companion satisfies

$$p^w B_\eta(ap + r) \equiv \eta(p) B_\eta(a) A(r) \pmod{p}.$$

Different character signatures therefore form separate Frobenius eigenspaces. Finally we apply the lifting calculus to a sporadic recurrence whose Apéry limit is one half of Catalan’s constant. A compact two-parameter product has as its mixed derivative a new χ_{-4} -harmonic closed form for the second solution. We prove the identity by an exact coupled holonomic reduction: two different recurrence residuals share an order-10 Ore factor, and finite coefficient propagation proves that their difference is 1. The formula then evaluates its own limit as $G/2$ by a positive-shell concentration argument, yields a positive error series for Catalan’s constant, and gives a character-twisted Lucas congruence on the uniform half-digit range. A second, Casoratian-based argument proves the unrestricted digit congruence at every prime for which the first-solution digit block has no zero modulo p . Finally, an endpoint binomial transform decouples the companion recurrence and proves the sharp global denominator bound $\text{lcm}(1, \dots, n)^2 B_E(n) \in \mathbb{Z}$. The same transform represents G as the finite part of a zero-balanced reciprocal-central- binomial series; the full transform family has an optimally accelerated member with convergence factor $1/3$, but also gives a sharp obstruction to obtaining irrationality from any fixed integral binomial transform. More generally, endpoint descent gives a complete denominator classification for the dilation family: its normalized companion has the L_n^2 bound at every index if and only if the dilation parameter is divisible by 4.

1 Introduction

Finite binomial sums with harmonic weights occur in Apéry limits, congruences, Feynman-integral calculations, and symbolic summation. A typical object is

$$B(n) = \sum_k S(n, k) W(n, k), \tag{1}$$

where S is a proper hypergeometric term and W is a polynomial in letters such as H_k , $H_{n-k}^{(2)}$, or

$$H_m^{(q,\chi)} := \sum_{1 \leq j \leq m} \frac{\chi(j)}{j^q}.$$

The standard direction is to introduce parameters into factorials and differentiate. The reverse direction is usually performed by inspired guessing: given W , one searches for a deformation that happens to produce it. The first purpose of this paper is to make that reverse step canonical.

Three principles emerge.

1. A harmonic letter is a *connected parameter derivative*. Boolean-lattice inclusion–exclusion removes every disconnected product of lower-weight letters.
2. Cyclotomic letters become ordinary hypergeometric parameters after splitting the relevant indices into finitely many residue classes.
3. Under Lucas factorisation, harmonic weight is a Frobenius degree: a letter of weight q contributes p^{-q} , while its Dirichlet character contributes the eigenvalue $\chi(p)$.

These statements sit next to, but are different from, several established theories. Difference-ring algorithms handle very general nested and cyclotomic sums [10, 2]; the point here is an elementary finite construction that lands directly in proper hypergeometric creative telescoping. Parameter deformations are central in the work of Straub–Zudilin on binomial powers and Apéry limits [11]. Taylor coefficients also occur in refined Lucas congruences for Apéry numbers [9]. Our focus is the inverse lifting operation and the character-resolved coefficientwise descent of an arbitrary homogeneous harmonic weight.

Parameter differentiation as a source of harmonic identities is classical and remains active; representative references include derivative-operator telescoping [14], all-order hypergeometric parameter expansions [4], and the recent monoidal-alphabet organisation of generalized harmonic sums [8]. We have not found in these sources the Boolean connected lift (6), the resulting arbitrary-weight inverse theorem, or the character-resolved Lucas theorem in the form proved below. This is a literature-search boundary, not a claim that no equivalent formulation can exist.

The last section gives a concrete test. The recurrence

$$(n+1)^2 u_{n+1} = (12n^2 + 12n + 4)u_n - 32n^2 u_{n-1} \quad (2)$$

has a binomial first solution and a second-solution limit involving Catalan’s constant. The lifting calculus explains, in one formula, both the product of weight-one letters and the apparently exceptional weight-two correction in a newly found closed form. Existing Lucas and Gessel–Lucas results for the sporadic families concern their integral first solutions [6, 12]. We have not found in the sources searched the finite χ_{-4} -harmonic second solution below or a twisted Lucas law for it; as above, this records the boundary of our search rather than asserting absolute priority.

Status convention. Every theorem and lemma below is proved. The Catalan recurrence proof is computer-assisted but exact: its scalar recurrences, right divisions over $\mathbb{Q}(x)$, leading coefficient polynomials, and finite propagation bounds are reproduced below and in the accompanying scripts. Irrationality of Catalan’s constant remains open; no transcendence or irrationality conclusion is asserted.

2 Harmonic alphabets and parameter products

For $m, q \geq 0$ put $H_0^{(q)} = 0$ and, for $q \geq 1$,

$$H_m^{(q)} = \sum_{j=1}^m j^{-q}.$$

If χ is periodic, define $H_m^{(q,\chi)}$ by the same formula with numerator $\chi(j)$. We give a letter $H^{(q,\chi)}$ weight q ; a product has total weight the sum of its letter weights.

The basic parameter product is

$$X_m(t) := \prod_{j=1}^m (1 - t/j)^{-1} = \frac{m!}{(1-t)_m}. \quad (3)$$

Its logarithm is the harmonic generating series

$$\log X_m(t) = \sum_{q \geq 1} \frac{H_m^{(q)}}{q} t^q. \quad (4)$$

For a residue class $a \bmod M$ we similarly set

$$X_{m;a,M}(t) := \prod_{\substack{1 \leq j \leq m \\ j \equiv a \pmod{M}}} (1 - t/j)^{-1}. \quad (5)$$

Then $\partial_t^q \log X_{m;a,M}(0) = (q-1)!$ times the harmonic sum restricted to that residue class.

2.1 The connected lift

Let $[r] = \{1, \dots, r\}$, write $t_T = \sum_{i \in T} t_i$, and take $X_m(0) = 1$.

Definition 2.1 (connected lift). For $r \geq 1$ define

$$\mathcal{R}_{m,r}(t_1, \dots, t_r) := \prod_{T \subseteq [r]} X_m(t_T)^{(-1)^{r-|T|}}. \quad (6)$$

For example,

$$\mathcal{R}_{m,1}(u) = X_m(u), \quad \mathcal{R}_{m,2}(u, v) = \frac{X_m(u+v)}{X_m(u)X_m(v)}.$$

The next theorem is the elementary engine of the paper.

Theorem 2.2 (single-letter inverse lift). *For every $m \geq 0$ and $r \geq 1$,*

$$\partial_{t_1} \cdots \partial_{t_r} \mathcal{R}_{m,r}(t_1, \dots, t_r)|_{t=0} = (r-1)! H_m^{(r)}. \quad (7)$$

Moreover, every nonconstant monomial of $\log \mathcal{R}_{m,r}$ contains every one of the variables $t_1, \dots, t_r$.

Proof. From (4),

$$\log \mathcal{R}_{m,r} = \sum_{q \geq 1} \frac{H_m^{(q)}}{q} \sum_{T \subseteq [r]} (-1)^{r-|T|} t_T^q.$$

If a monomial omits t_i , its coefficient cancels between T and $T \cup \{i\}$. Consequently every surviving monomial contains all r variables and has degree at least r . The coefficient of $t_1 \cdots t_r$ comes only from $q = r$; in t_T^r it is zero unless $T = [r]$, when it is $r!$. Thus the full mixed derivative of $\log \mathcal{R}_{m,r}$ at the origin is $(r-1)! H_m^{(r)}$. Since the logarithm has no terms of degree below r , the same mixed derivative of its exponential $\mathcal{R}_{m,r}$ has the same value. $\square$

This is a finite-difference version of the moment–cumulant relation. The Boolean signs are exactly what prevents, for example, H_m^2 from contaminating the lift of $H_m^{(2)}$.

Theorem 2.3 (inverse lifting of a harmonic polynomial). *Let W be a polynomial over a characteristic-zero field in finitely many letters $H_{m_i}^{(q)}$, homogeneous of total weight w . Then W is a finite linear combination of order- w mixed derivatives at the origin of products and quotients of terms $X_{m_i}(L(t))$, where L is a linear form with coefficients 0 or 1.*

The construction is explicit: replace each occurrence of $H_m^{(q)}$ by $\mathcal{R}_{m,q}/(q-1)!$, give different occurrences disjoint parameter blocks, multiply, and differentiate once in every parameter.

Proof. It suffices by linearity to treat a monomial $\prod_{i=1}^s H_{m_i}^{(q_i)}$ with $\sum_i q_i = w$. Give its i th factor a fresh block of q_i variables and multiply the connected lifts $\mathcal{R}_{m_i,q_i}/(q_i-1)!$. The variables are disjoint, so the order- w derivative factors. Theorem 2.2 gives the required monomial. $\square$

Corollary 2.4 (economical weight two). *Every quadratic harmonic expression in the letters $h_i = H_{m_i}$ and $q_i = H_{m_i}^{(2)}$ is a linear combination of mixed derivatives of*

$$X_{m_i}(u)X_{m_j}(v), \quad \frac{X_{m_i}(u+v)}{X_{m_i}(u)X_{m_i}(v)}. \quad (8)$$

Indeed the first block gives $h_i h_j$ and the second gives q_i .

Remark 2.5 (polarisation). If a two-parameter term $F(u, v)$ is expensive to telescope, its mixed derivative can be recovered from three one-parameter families:

$$F_{uv}(0, 0) = \frac{1}{2} \left(\frac{d^2}{dt^2} F(t, t) \Big|_{t=0} - F_{uu}(0, 0) - F_{vv}(0, 0) \right). \quad (9)$$

This elementary identity was effective in the binomial companion calculations that motivated the present formalism.

3 Cyclotomic letters and finite residue splitting

Let $M \geq 1$. Every periodic weight χ of period M decomposes into residue indicators, so

$$H_m^{(q,\chi)} = \sum_{a=1}^M \chi(a) \sum_{\substack{1 \leq j \leq m \\ j \equiv a \pmod{M}}} j^{-q}. \quad (10)$$

Applying Theorem 2.2 to $X_{m;a,M}$ therefore gives an inverse lift for every cyclotomic letter. The construction is over the field generated by the values of χ .

The relevance to hypergeometric summation is that residue products are Pochhammer ratios. Write $m = M\ell + s$, $0 \leq s < M$. Up to a factor independent of ℓ ,

$$X_{M\ell+s;a,M}(t) = \frac{(a/M)_{\ell+\epsilon_{a,s}}}{((a-t)/M)_{\ell+\epsilon_{a,s}}}, \quad \epsilon_{a,s} \in \{0, 1\}, \quad (11)$$

with the evident convention for the representative $a \in \{1, \dots, M\}$. Its quotient under $\ell \mapsto \ell + 1$ is rational in ℓ and t .

Theorem 3.1 (cyclotomic inverse lift). *Let W be a polynomial in finitely many cyclotomic harmonic letters with a common period dividing M . After splitting each affine harmonic argument and each summation index modulo M , W is a finite linear combination of parameter derivatives of products of Pochhammer ratios whose shift quotients are rational.*

Proof. Use (10), Theorem 2.3, and then (11). Only finitely many residue choices occur. On each choice, floors and the corrections $\epsilon_{a,s}$ are fixed affine functions, so all shift quotients are rational. $\square$

For a real primitive character one may package the residue factors more compactly as

$$Y_m^\chi(t) := \prod_{j=1}^m (1 - t/j)^{-\chi(j)}, \quad \log Y_m^\chi(t) = \sum_{q \geq 1} \frac{H_m^{(q, \chi)}}{q} t^q. \quad (12)$$

When $\chi(j) \in \{0, \pm 1\}$ this itself has rational shift quotients after residue splitting. For non-real characters the residue-indicator construction is preferable: it keeps every hypergeometric exponent integral.

4 Holonomicity is automatic

We use “proper hypergeometric” in the standard Wilf–Zeilberger sense: the quotients under unit shifts of the discrete variables are rational, with finite factorial support.

Theorem 4.1 (cyclotomic harmonic holonomicity). *Let $S(n, k)$ be a proper hypergeometric term with finite, linearly bounded support in k . Let $W(n, k)$ be a polynomial in cyclotomic harmonic letters whose arguments are nonnegative affine integer forms on that support. Then*

$$B(n) = \sum_k S(n, k) W(n, k) \quad (13)$$

is P-recursive over the number field generated by the coefficients and character values. Moreover, a recurrence and certificate for B can be obtained by finitely many ordinary proper-hypergeometric creative-telescoping computations followed by linear elimination.

Proof. Choose a common period M and apply Theorem 3.1. Split n, k modulo M . Each resulting lift is a proper hypergeometric term over a rational function field in its parameters. Zeilberger’s theorem supplies a recurrence for its finite definite sum [13, 7]. Differentiate the recurrence to the finite order prescribed by W and set the parameters to zero. A derivative of a parameter-dependent recurrence may couple the target derivative to lower derivatives, but there are only finitely many of them. Closure of holonomic systems under finite elimination gives a scalar recurrence for the desired derivative. Finally interlace the finitely many residue subsequences; P-recursive sequences are closed under finite interlacing. $\square$

Remark 4.2 (what the theorem does and does not optimize). Theorem 4.1 guarantees an algorithm, not a minimal recurrence. Direct difference-field summation can be dramatically smaller, and parameter families may produce large nonminimal operators before elimination. The advantage is universality: failure of a direct harmonic Gosper ansatz is never a failure of holonomicity.

Corollary 4.3 (bounded lifting complexity). *If W is a sum of N harmonic monomials of total weight w , the construction needs at most N order- w parameter derivatives. At weight two, one may instead use exactly one block of (8) per nonzero quadratic coefficient. Thus parameter lifting has complexity controlled by the support of W , not by the sizes of its harmonic arguments.*

5 Character-resolved Lucas descent

Fix a prime p . In this section all χ are Dirichlet characters (the trivial character is allowed). Congruences involving rational harmonic numbers are interpreted in $\mathbb{Z}_{(p)}$ after the displayed power of p has cleared the possible p -denominators. Assume throughout this section that p does not divide the periods of the characters.

Lemma 5.1 (one-letter descent). *Let $q \geq 1$, let χ be completely multiplicative, and let $y \geq 0$. Then*

$$p^q H_y^{(q, \chi)} = \chi(p) H_{\lfloor y/p \rfloor}^{(q, \chi)} + p^q T, \quad T = \sum_{\substack{j \leq y \\ p \nmid j}} \frac{\chi(j)}{j^q} \in \mathbb{Z}_{(p)}. \quad (14)$$

Proof. Separate the summands with $p \mid j$, write $j = p\ell$, and use $\chi(p\ell) = \chi(p)\chi(\ell)$. The complementary denominators are p -units. $\square$

A character signature is a product $\eta = \prod_i \chi_i$, with repetitions allowed. A monomial $\prod_i H_{x_i}^{(q_i, \chi_i)}$ has weight $w = \sum_i q_i$ and signature η .

Lemma 5.2 (monomial descent). *Suppose $0 \leq y_i < p^2$ for every i . Then*

$$p^w \prod_i H_{y_i}^{(q_i, \chi_i)} \equiv \eta(p) \prod_i H_{\lfloor y_i/p \rfloor}^{(q_i, \chi_i)} \pmod{p}. \quad (15)$$

Both sides belong to $\mathbb{Z}_{(p)}$.

Proof. Multiply the identities (14). Because $\lfloor y_i/p \rfloor < p$, every descended denominator is a p -unit. Every term other than the product of the leading terms contains a positive power of p . $\square$

We now state a strong, clean digit theorem. It is designed so that no informal “surviving rectangle” argument is needed.

Theorem 5.3 (character-resolved harmonic Lucas theorem). *Let $S(n, k) \in \mathbb{Z}$, extended by zero outside its finite support, and put $A(n) = \sum_k S(n, k)$. Let*

$$W = \sum_{\eta} W_{\eta}$$

be a finite sum in which each $W_{\eta}(n, k)$ is a homogeneous weight- w polynomial in cyclotomic harmonic letters of signature η . Put $B_{\eta}(n) = \sum_k S(n, k) W_{\eta}(n, k)$.

Let $n = ap + r$ with $0 \leq a, r < p$. Assume:

(L1) *for every $b \geq 0$ and $0 \leq s < p$,*

$$S(ap + r, bp + s) \equiv S(a, b) S(r, s) \pmod{p};$$

(L2) *every harmonic argument x occurring in W satisfies $0 \leq x(ap + r, bp + s) < p^2$ on the support and*

$$\left\lfloor \frac{x(ap + r, bp + s)}{p} \right\rfloor = x(a, b);$$

(L3) *the coefficients of W belong to $\mathbb{Z}_{(p)}$.*

Then

$$p^w B_{\eta}(ap + r) \equiv \eta(p) B_{\eta}(a) A(r) \pmod{p} \quad (16)$$

for every signature η , and hence

$$p^w B(ap + r) \equiv A(r) \sum_{\eta} \eta(p) B_{\eta}(a) \pmod{p}. \quad (17)$$

Proof. Write each k uniquely as $bp + s$, $0 \leq s < p$, and enlarge to a full finite block using the zero extension of S . Lemma 5.2, linearity, and (L2) give

$$p^w W_\eta(ap + r, bp + s) \equiv \eta(p) W_\eta(a, b) \pmod{p}.$$

This expression is p -integral on every cell. Therefore (L1) gives

$$\begin{aligned} p^w B_\eta(ap + r) &\equiv \eta(p) \sum_{b,s} S(a, b) S(r, s) W_\eta(a, b) \pmod{p} \\ &= \eta(p) B_\eta(a) A(r), \end{aligned}$$

which proves (16); summing over signatures gives (17). $\square$

Corollary 5.4 (Frobenius eigenspaces). *If all monomials have one signature η , the companion has the scalar digit law (16). If several signatures occur, their components do not mix: the digit map acts diagonally with eigenvalues $\eta(p)$. Thus character-homogeneity is not a technical convenience but the condition that the harmonic jet lie in one Frobenius eigenspace.*

Corollary 5.5 (coefficientwise jet descent). *Collect all homogeneous harmonic polynomials of weight w satisfying (L2) into a formal weight- w jet $\mathcal{J}_w(n)$. Then the digit map on $\mathcal{J}_w$ is the tensor product of multiplication by $A(r)$ with the diagonal character action $\eta \mapsto \eta(p)$, together with the Tate factor p^{-w} .*

Remark 5.6 (scope). The restriction $a < p$ ensures that descended harmonic numbers are p -integral. Iterating (16) without rechecking this point is not justified. Multi-digit laws require either stronger denominator control or a divided-power formulation. The theorem is deliberately single-digit.

6 A compact lift for the Catalan companion

Let χ_{-4} be the real character modulo 4:

$$\chi_{-4}(j) = 0 \ (2 \mid j), \quad \chi_{-4}(j) = (-1)^{(j-1)/2} \ (j \text{ odd}),$$

and abbreviate $K_m^{(q)} = H_m^{(q, \chi_{-4})}$. Define

$$S_E(n, k) = \binom{n}{k} \binom{2k}{k} \binom{2n-2k}{n-k}, \quad A_E(n) = \sum_{k=0}^n S_E(n, k). \quad (18)$$

The sequence $A_E = 1, 4, 20, 112, 676, \dots$ satisfies (2). The second solution normalized by $B_E(0) = 0$, $B_E(1) = 1$ has Apéry limit $G/2$, where $G = L(2, \chi_{-4})$ is Catalan's constant; this value belongs to the established sporadic Apéry-limit tables [15, 3, 6]. This A_E is the scaled Catalan–Larcombe–French sequence. A useful second shell is

$$A_E(n) = \sum_{0 \leq k \leq n/2} \binom{2k}{k}^2 \binom{n}{2k} 4^{n-2k}, \quad (19)$$

and its first-solution congruences have been studied extensively [5, 12]. The arithmetic results below instead concern its rational second solution.

Theorem 6.1 (Catalan harmonic companion). *For every $n \geq 0$, the normalized second solution of (2) is*

$$B_E(n) = \sum_{k=0}^n S_E(n, k) \left\{ \frac{1}{2} K_{2k}^{(2)} + \left(\frac{3}{4} H_k - \frac{1}{2} H_{2k} \right) \left(K_{2k}^{(1)} - K_{2n-2k}^{(1)} \right) \right\}. \quad (20)$$

The first values are

$$0, 1, 7, \frac{404}{9}, \frac{2603}{9}, \frac{428284}{225}, \frac{2884316}{225}.$$

These values satisfy (2) for every $n \geq 1$. Moreover,

$$\lim_{n \rightarrow \infty} \frac{B_E(n)}{A_E(n)} = \frac{G}{2}. \quad (21)$$

The formula was originally found by exact linear algebra in a weight-two cyclotomic alphabet. The following proposition explains why all three pieces of its weight occur together. The recurrence proof is completed after the generating-function factorisation below.

Proposition 6.2 (two-parameter Catalan lift). *Let X be (3), let $Y_m(t) = Y_m^{X^{-4}}(t)$ be (12), and put*

$$\begin{aligned} \mathcal{F}_E(n, k; u, v) := S_E(n, k) & \frac{X_k(3u/4)}{X_{2k}(u/2)} \frac{Y_{2k}(v)}{Y_{2k}(u+v/2)} \\ & \times \frac{Y_{2n-2k}(v)}{Y_{2k}(u)Y_{2k}(v/2)}. \end{aligned} \quad (22)$$

Then the summand in (20) is exactly

$$\partial_u \partial_v \mathcal{F}_E(n, k; u, v)|_{u=v=0}. \quad (23)$$

Proof. All parameter factors equal 1 at the origin. From (4) and (12), the first logarithmic derivatives of (22) are

$$\partial_u \log \mathcal{F}_E|_0 = \frac{3}{4}H_k - \frac{1}{2}H_{2k}, \quad \partial_v \log \mathcal{F}_E|_0 = K_{2k}^{(1)} - K_{2n-2k}^{(1)}.$$

Only the last quotient in (22) has a mixed logarithmic derivative. It is the connected lift $Y_{2k}(u+v/2)/(Y_{2k}(u)Y_{2k}(v/2))$, so

$$\partial_u \partial_v \log \mathcal{F}_E|_0 = \frac{1}{2}K_{2k}^{(2)}.$$

Finally $F_{uv}/F = (\log F)_u(\log F)_v + (\log F)_{uv}$, which is precisely the weight in (20). $\square$

The lift is not merely formal: it reduces to proper hypergeometric terms.

Lemma 6.3 (parity reduction). *Put $P_q(t) = Y_{2q}(t)$. For $j \geq 0$,*

$$P_{2j}(t) = \frac{(1/4)_j}{((1-t)/4)_j} \frac{((3-t)/4)_j}{(3/4)_j}, \quad (24)$$

$$P_{2j+1}(t) = P_{2j}(t) \frac{4j+1}{4j+1-t}. \quad (25)$$

Proof. In Y_{4j} the positive character values occur at $1, 5, \dots, 4j-3$ and the negative ones at $3, 7, \dots, 4j-1$. Grouping these two products and extracting powers of 4 gives (24). Passing from $4j$ to $4j+2$ adds only the positive factor $(1-t/(4j+1))^{-1}$, giving (25). $\square$

Corollary 6.4 (certifiability of the Catalan formula). *After writing $n = 2m + \epsilon$ and $k = 2j + \delta$, the summand $\mathcal{F}_E(n, k; u, v)$ is proper hypergeometric in m, j on each of the four parity cells. Consequently the right side of (20) is P -recursive and admits a finite creative-telescoping certificate.*

Proof. The X -factors are factorial ratios. Lemma 6.3 turns every Y -factor into a Pochhammer ratio after the parity choice. The remaining factor S_E is proper hypergeometric. Apply Theorem 4.1. $\square$

There is also a useful convolution structure. With $\ell = n - k$, (22) factors as

$$\binom{n}{k} P_k(u, v) Q_\ell(v),$$

where

$$P_k(u, v) = \binom{2k}{k} \frac{X_k(3u/4)}{X_{2k}(u/2)} \frac{Y_{2k}(v) Y_{2k}(u + v/2)}{Y_{2k}(u) Y_{2k}(v/2)},$$

$$Q_\ell(v) = \binom{2\ell}{\ell} Y_{2\ell}(v)^{-1}.$$

Thus the exponential generating function of the lifted sum is the product

$$\left(\sum_{k \geq 0} P_k(u, v) \frac{x^k}{k!} \right) \left(\sum_{\ell \geq 0} Q_\ell(v) \frac{x^\ell}{\ell!} \right). \quad (26)$$

The determinant hidden in this factorisation leads to a substantially smaller proof than the raw parity-split telescoper. Put

$$c_k = \binom{2k}{k}, \quad \alpha_k = \frac{3}{4}H_k - \frac{1}{2}H_{2k}, \quad a_k = c_k \alpha_k, \quad d_k = c_k K_{2k}^{(1)}, \quad (27)$$

and

$$e_k = c_k \left(\frac{1}{2} K_{2k}^{(2)} + \alpha_k K_{2k}^{(1)} \right). \quad (28)$$

If capital letters denote exponential generating functions, binomial convolution gives

$$B(x) := \sum_{n \geq 0} B_E(n) \frac{x^n}{n!} = E(x)C(x) - A(x)D(x). \quad (29)$$

Thus the companion is a 2×2 determinant of four harmonic jets.

The elementary increments behind the certificate are

$$\alpha_{k+1} - \alpha_k = \frac{k}{2(k+1)(2k+1)},$$

$$K_{2k+2}^{(1)} - K_{2k}^{(1)} = \frac{(-1)^k}{2k+1}, \quad K_{2k+2}^{(2)} - K_{2k}^{(2)} = \frac{(-1)^k}{(2k+1)^2}. \quad (30)$$

With $p_k = (-1)^k c_k$ and $s_k = (-1)^k a_k$, direct substitution gives the coupled system

$$\begin{aligned} (k+1)c_{k+1} &= 2(2k+1)c_k, \\ (k+1)^2 a_{k+1} &= 2(k+1)(2k+1)a_k + k c_k, \\ (k+1)d_{k+1} &= 2(2k+1)d_k + 2p_k, \\ (k+1)p_{k+1} &= -2(2k+1)p_k, \\ (k+1)^2 s_{k+1} &= -2(k+1)(2k+1)s_k - k p_k, \\ (k+1)^2 e_{k+1} &= 2(k+1)(2k+1)e_k + p_k + 2(k+1)s_k + k d_k. \end{aligned} \quad (31)$$

In particular, the scalar input recurrences below are consequences of first-order rational identities, not interpolated recurrences.

Let $\theta = x D_x$ and define

$$\mathcal{L} = (\theta + 1)^2 D_x - (12\theta^2 + 12\theta + 4) + 32x(\theta + 1). \quad (32)$$

For an exponential generating function $F(x) = \sum f_n x^n / n!$, the coefficient of $x^n / n!$ in $\mathcal{L}F$ is

$$(n+1)^2 f_{n+1} - (12n^2 + 12n + 4)f_n + 32n^2 f_{n-1}.$$

Set $Y_1 = \mathcal{L}(EC)$ and $Y_2 = \mathcal{L}(AD)$.

Proposition 6.5 (exact common-factor certificate). *The two residual series satisfy*

$$Y_1 - Y_2 = 1. \quad (33)$$

Exact holonomic closure applied to (31) produces annihilators $\mathcal{H}_1, \mathcal{H}_2$ of orders 44, 29. After clearing scalar denominators, both have the same right factor $\mathcal{C} \in \mathbb{Q}[x]\langle D_x \rangle$, of order 10 and degree 29:

$$\mathcal{H}_1 = \mathcal{U}_1 \mathcal{C}, \quad \mathcal{H}_2 = \mathcal{U}_2 \mathcal{C}, \quad \text{ord}(\mathcal{U}_1) = 34, \quad \text{ord}(\mathcal{U}_2) = 19. \quad (34)$$

Moreover $\mathcal{C}(1) = 0$.

Proof. All operations are exact over $\mathbb{Q}(x)$. Elimination from (31), exponential normalisation, product closure, and association by (32) give $\mathcal{H}_1$ and $\mathcal{H}_2$. An independently fitted order-11, degree-17 operator is used only to locate $\mathcal{C}$; the identities in (34) are then established by exact zero-remainder right division, so the fit has no logical role in the certificate.

We record the finite uniqueness check. For a polynomial differential operator $T = \sum t_{j,i} x^i D_x^j$, the coefficient of f_{n+r} in $T(\sum f_n x^n)$, where $r = \max(j - i)$, is

$$P_T(n) = \sum_{j-i=r} t_{j,i} (n - i + 1) \cdots (n - i + j). \quad (35)$$

For $\mathcal{U}_1, \mathcal{U}_2, \mathcal{C}$, the largest forward shift is 4 and, up to nonzero rational constants, the three forward coefficients are

$$\begin{aligned} & (z+1)(z+2)^2(z+3)^2(z+4)^3(z+5)^3 \prod_{j=6}^9 (z+j)^2 \\ & \times \left(z^5 + \frac{61}{2}z^4 + 370z^3 + \frac{4461}{2}z^2 + \frac{26709}{4}z + \frac{63465}{8} \right)^3, \\ & (z+1)(z+2)^2(z+3)^2(z+4)^3(z+5)^3 \prod_{j=6}^9 (z+j)^2, \\ & (z+1)(z+2)^2(z+3)^2(z+4)^4(z+5). \end{aligned}$$

They are nonzero at every nonnegative integer. Their complete forward coefficients are present from $z = 30, 15, 6$, respectively. Exact expansion gives the first 34 coefficients of $\mathcal{C}Y_1$ and the first 19 coefficients of $\mathcal{C}Y_2$ equal to zero. The first two recurrences therefore imply $\mathcal{C}Y_1 = \mathcal{C}Y_2 = 0$. Since $\mathcal{C}(1) = 0$, the series $Y_1 - Y_2 - 1$ is a $\mathcal{C}$ -solution; its first 10 coefficients vanish, and the third recurrence gives (33).

The reconstructed operator $\mathcal{C}$ has SHA-256 fingerprint

5f255471c10932d5a24a7cf4f429729bda8d9f73740aa3a1c37e261e70312fba.

The accompanying exact-arithmetic scripts reconstruct it and check every assertion. □

Proof of Theorem 6.1, recurrence part. By (29), Proposition 6.5 says $\mathcal{L}B = 1$. Coefficient comparison gives (2) for $n \geq 1$. The finite sum gives $B_E(0) = 0$ and $B_E(1) = 1$, hence it is the normalized second solution. □

The harmonic representation also evaluates its own Apéry limit, without appealing to the sporadic tables. This is one of the principal benefits of having a finite companion formula.

Proposition 6.6 (direct Catalan limit). *Formula (20) implies (21) by elementary estimates.*

Proof. Write $D_m = K_{2m}^{(1)}$, $Q_m = K_{2m}^{(2)}$, and put $\mu_n(k) = S_E(n, k)/A_E(n)$. Since $S_E(n, k) = S_E(n, n - k)$, symmetrising (20) gives, with $\ell = n - k$,

$$\frac{B_E(n)}{A_E(n)} = \sum_{k=0}^n \mu_n(k) \left\{ \frac{1}{4}(Q_k + Q_\ell) + \frac{1}{2}(\alpha_k - \alpha_\ell)(D_k - D_\ell) \right\}. \quad (36)$$

We first control the shell measure. The elementary bounds

$$\binom{2m}{m} \leq 4^m, \quad \binom{2m}{m} \geq \frac{4^m}{2m+1}, \quad \max_k \binom{n}{k} \geq \frac{2^n}{n+1}$$

give, by taking $k = \lfloor n/2 \rfloor$,

$$A_E(n) \geq \frac{8^n}{(n+1)^3}. \quad (37)$$

On the other hand, $S_E(n, k) \leq 4^n \binom{n}{k}$. The exponential-moment estimate

$$\sum_{k \leq n/4} \binom{n}{k} \leq 3^{n/4} \sum_{k=0}^n \binom{n}{k} 3^{-k} = \frac{4^n}{3^{3n/4}}$$

and symmetry therefore show that

$$\sum_{k < n/4 \text{ or } k > 3n/4} \mu_n(k) \leq 2(n+1)^3 \left(\frac{2}{3^{3/4}} \right)^n. \quad (38)$$

On the bulk $n/4 \leq k, \ell \leq 3n/4$, the alternating-series estimate gives

$$|Q_m - G| \leq (2m+1)^{-2}, \quad |D_k - D_\ell| \leq (2 \min(k, \ell) + 1)^{-1}. \quad (39)$$

Moreover (30) implies

$$0 \leq \alpha_{m+1} - \alpha_m \leq \frac{1}{4(m+1)},$$

so $|\alpha_k - \alpha_\ell|$ is bounded uniformly on the bulk. Thus the expression in braces in (36) is $G/2 + O(1/n)$ there. Globally, $|D_m|, |Q_m| \leq 1$ and $0 \leq \alpha_m \leq H_m/4$, so the same expression is $O(\log(n+1))$. Equation (38) makes the boundary contribution tend to zero, proving (21). $\square$

Completion of the proof of Theorem 6.1. The recurrence and initial values were proved above, and Proposition 6.6 proves the claimed limit. $\square$

Corollary 6.7 (positive error formula). *The approximants $B_E(n)/A_E(n)$ increase strictly to $G/2$, and*

$$\frac{G}{2} - \frac{B_E(n)}{A_E(n)} = \sum_{m=n+1}^{\infty} \frac{32^{m-1}}{m^2 A_E(m) A_E(m-1)}, \quad (40)$$

$$G = 2 \sum_{m=1}^{\infty} \frac{32^{m-1}}{m^2 A_E(m) A_E(m-1)}. \quad (41)$$

Proof. For two solutions A, B of (2), their Casoratian $W_n = A_n B_{n-1} - A_{n-1} B_n$ satisfies

$$W_{n+1} = \frac{32n^2}{(n+1)^2} W_n.$$

The initial values give $W_1 = -1$, hence $W_n = -32^{n-1}/n^2$. Dividing by $A_E(n)A_E(n-1) > 0$ yields

$$\frac{B_E(n)}{A_E(n)} - \frac{B_E(n-1)}{A_E(n-1)} = \frac{32^{n-1}}{n^2 A_E(n) A_E(n-1)} > 0.$$

Sum this identity and use Proposition 6.6; taking $n = 0$ gives (41). $\square$

The error formula does not expose the true denominator: its summands contain quotients of first solutions. The harmonic formula gives a first bound.

Corollary 6.8 (termwise denominator bound). *Put $L_0 = 1$ and $L_m = \text{lcm}(1, 2, \dots, m)$ for $m \geq 1$. Then*

$$4L_{2n}^2 B_E(n) \in \mathbb{Z}. \quad (42)$$

Proof. Every first-order harmonic letter in (20) has denominator dividing L_{2n} , and every second-order letter has denominator dividing L_{2n}^2 . The factor 4 clears the rational coefficients in α_k and the leading $1/2$; the shell is integral. $\square$

The factor L_{2n}^2 is not sharp. A binomial transform at the endpoint 4 of the two characteristic roots exposes all the cancellation that is invisible termwise in (20).

Theorem 6.9 (sharp global denominator). *For every $n \geq 0$,*

$$L_n^2 B_E(n) \in \mathbb{Z}. \quad (43)$$

More precisely, define the endpoint transform

$$T_n = \sum_{k=0}^n \binom{n}{k} (-4)^{n-k} B_E(k). \quad (44)$$

Then $T_0 = 0$ and, with $T_{-1} = 0$,

$$(n+1)^2 T_{n+1} - 16n^2 T_{n-1} = (-4)^n \quad (n \geq 0). \quad (45)$$

For $n \geq 1$ this has the signed sum-of-squares solution

$$(-1)^{n-1} T_n = \sum_{\substack{0 \leq j < n \\ n-j \text{ modd}}} R_{n,j}^2, \quad R_{n,j} = 2^{n-1} \frac{(j+2)(j+4) \cdots (n-1)}{(j+1)(j+3) \cdots n}, \quad (46)$$

where an empty numerator is 1. Every $L_n R_{n,j}$ is an integer.

Proof. Let $B(z) = \sum_{n \geq 0} B_E(n) z^n$ and $\theta = z D_z$. Including the exceptional initial residual $B_E(1) - 4B_E(0) = 1$, the recurrence is equivalent to

$$\{\theta^2 - z(12\theta^2 + 12\theta + 4) + 32z^2(\theta + 1)^2\} B(z) = z. \quad (47)$$

The ordinary generating function of (44) is

$$T(z) = \frac{1}{1+4z} B\left(\frac{z}{1+4z}\right). \quad (48)$$

A chain-rule substitution in (47) gives

$$\{\theta^2 - 16z^2(\theta + 1)^2\} T(z) = \frac{z}{1+4z}.$$

Coefficient comparison is exactly (45).

Iterating (45) separately on the even and odd indices shows that the contribution of its forcing term $(-4)^j$ to T_n , for $n - j$ odd, is

$$\frac{(-4)^j}{(j+1)^2} \prod_{\substack{j+2 \leq r < n \\ r \equiv j \pmod{2}}} \frac{16r^2}{(r+1)^2} = (-1)^j R_{n,j}^2.$$

All admissible j have parity $n - 1$, proving (46).

It remains to prove the asserted denominator of $R_{n,j}$. Fix an odd prime p . For every $a \geq 1$, the multiples of p^a in the interval $[j+1, n]$ are p^a times a consecutive interval of integers. Because p^a is odd, multiplication by it preserves parity. Hence the number of those multiples in the denominator progression of $R_{n,j}$ exceeds the number in its numerator progression by at most one. Summing over a gives

$$-v_p(R_{n,j}) \leq \#\{a : p^a \leq n\} = v_p(L_n)$$

whenever the left side is positive. At $p = 2$, the denominator progression is either odd or has 2-adic valuation at most $v_2(n!) \leq n - 1$, which is absorbed by the factor 2^{n-1} . Thus $L_n R_{n,j} \in \mathbb{Z}$.

Equation (46) now gives $L_n^2 T_n \in \mathbb{Z}$. Binomial inversion in (44) reads

$$B_E(n) = \sum_{k=0}^n \binom{n}{k} 4^{n-k} T_k.$$

Since $L_k \mid L_n$ for $k \leq n$, every summand becomes integral after multiplication by L_n^2 , proving (43). $\square$

Theorem 6.10 (general endpoint-denominator classification). *Let $b \in \mathbb{Z}$, and let $B_n^{(b)}$ be the unique rational sequence with $B_0^{(b)} = 0$, $B_1^{(b)} = 1$, and*

$$(n+1)^2 B_{n+1}^{(b)} = b(3n^2 + 3n + 1)B_n^{(b)} - 2b^2 n^2 B_{n-1}^{(b)} \quad (n \geq 1). \quad (49)$$

Define

$$T_n^{(b)} = \sum_{k=0}^n \binom{n}{k} (-b)^{n-k} B_k^{(b)}. \quad (50)$$

Then

$$(n+1)^2 T_{n+1}^{(b)} - b^2 n^2 T_{n-1}^{(b)} = (-b)^n \quad (n \geq 0), \quad (51)$$

and, for $n \geq 1$,

$$(-1)^{n-1} T_n^{(b)} = b^{n-1} \sum_{\substack{0 \leq j < n \\ n-j \text{ odd}}} \left\{ \frac{(j+2)(j+4) \cdots (n-1)}{(j+1)(j+3) \cdots n} \right\}^2. \quad (52)$$

Moreover, the following are equivalent:

$$4 \mid b \quad \Longleftrightarrow \quad L_n^2 B_n^{(b)} \in \mathbb{Z} \quad \text{for every } n \geq 0. \quad (53)$$

Proof. Let $B^{(b)}(z) = \sum_{n \geq 0} B_n^{(b)} z^n$. Including the initial residual $B_1^{(b)} - bB_0^{(b)} = 1$, recurrence (49) is equivalent to

$$\{\theta^2 - bz(3\theta^2 + 3\theta + 1) + 2b^2 z^2(\theta + 1)^2\} B^{(b)}(z) = z. \quad (54)$$

The binomial transform has generating function

$$T^{(b)}(z) = \frac{1}{1+bz} B^{(b)}\left(\frac{z}{1+bz}\right).$$

Substitution in (54) cancels both the middle term and every term of order z^3 , leaving

$$\{\theta^2 - b^2 z^2(\theta + 1)^2\} T^{(b)}(z) = \frac{z}{1+bz}.$$

Coefficient comparison proves (51).

As in the proof of Theorem 6.9, iterate the even and odd recurrences. The contribution of the forcing term with index j to $T_n^{(b)}$ is

$$\frac{(-b)^j}{(j+1)^2} \prod_{\substack{j+2 \leq r < n \\ r \equiv j \pmod{2}}} \frac{b^2 r^2}{(r+1)^2} = (-1)^j b^{n-1} \left\{ \frac{(j+2)(j+4) \cdots (n-1)}{(j+1)(j+3) \cdots n} \right\}^2.$$

All admissible j have parity $n-1$, proving (52).

Suppose first that $b = 4d$. Write the rational number in braces in (52) as $P_{n,j}$. The denominator argument in Theorem 6.9 proved the stronger statement

$$L_n 2^{n-1} P_{n,j} \in \mathbb{Z}.$$

Consequently

$$L_n^2 b^{n-1} P_{n,j}^2 = d^{n-1} (L_n 2^{n-1} P_{n,j})^2 \in \mathbb{Z}.$$

Equation (52) therefore gives $L_n^2 T_n^{(b)} \in \mathbb{Z}$. Binomial inversion reads

$$B_n^{(b)} = \sum_{k=0}^n \binom{n}{k} b^{n-k} T_k^{(b)}.$$

Since $L_k \mid L_n$, multiplying by L_n^2 makes every summand integral. This proves the forward implication in (53).

For the converse, direct propagation of (49) gives

$$B_4^{(b)} = \frac{2603b^3}{576}, \quad B_8^{(b)} = \frac{6802537507b^7}{180633600}. \quad (55)$$

Since $L_4 = 12$ and 2603 is odd, integrality at $n = 4$ says $2603b^3/4 \in \mathbb{Z}$, forcing b to be even. If $b = 2d$ with d odd, then $L_8 = 840$ and the second value in (55) gives

$$L_8^2 B_8^{(b)} = \frac{6802537507b^7}{256} = \frac{6802537507d^7}{2},$$

which is not integral. Thus integrality at every index forces $4 \mid b$, completing the reverse implication. $\square$

Remark 6.11 (why this family decouples). For the general quadratic recurrence and a nonzero shift h ,

$$(n+1)^2 u_{n+1} = \{\alpha n(n+1) + \beta\} u_n - \gamma n^2 u_{n-1},$$

the binomial shift conjugates its ordinary generating-function operator to

$$\begin{aligned} & \theta^2 - z\{(\alpha - 3h)(\theta^2 + \theta) + (\beta - h)\} \\ & + z^2(\gamma - 2\alpha h + 3h^2)(\theta + 1)^2 + z^3 h(\gamma - \alpha h + h^2)(\theta + 1)(\theta + 2). \end{aligned}$$

The coefficient at z vanishes only when $\alpha = 3h$ and $\beta = h$. Under these conditions the order-three coefficient vanishes exactly when $\gamma = 2h^2$. Thus, apart from the trivial zero-shift degeneracy, (49) is exactly the endpoint-decoupling locus, not an arbitrary parameter insertion. It is also a dilation: $B_n^{(b)} = b^{n-1} B_n^{(1)}$ for $n \geq 1$. The arithmetic content of Theorem 6.10 is that precisely the dilations $4 \mid b$ supply the full 2-adic factor needed by the parity quotient while the odd-prime deficit is absorbed by L_n .

Remark 6.12 (where the cancellation occurs). The summands of the harmonic formula (20) can retain primes between n and $2n$, so Theorem 6.9 is genuinely global in that shell. The endpoint transform moves the cancellation into the elementary parity interlacing of (46); there it is termwise after multiplication by L_n^2 .

The same transform retains the Apéry limit in an unexpected way: G becomes the finite part of a logarithmically divergent reciprocal-binomial sum. Reciprocal-binomial accelerations of G have a substantial history [1]; the point here is that this particular zero-balanced finite part is forced directly by the endpoint transform of the companion.

Corollary 6.13 (endpoint finite part and a series for G). *One has the absolutely convergent identity*

$$\sum_{j=1}^{\infty} \left\{ \frac{16^{j-1}}{j^2 \binom{2j}{j}^2} - \frac{\pi}{16j} \right\} = \frac{\pi}{4} \log 2 - \frac{G}{2}. \quad (56)$$

Equivalently,

$$G = \frac{\pi}{2} \log 2 + \sum_{j=1}^{\infty} \left\{ \frac{\pi}{8j} - \frac{2 \cdot 16^{j-1}}{j^2 \binom{2j}{j}^2} \right\}. \quad (57)$$

In particular, the even part of the rational endpoint transform satisfies

$$\lim_{m \rightarrow \infty} \left\{ \frac{T_{2m}}{\binom{2m}{m}^2} + \frac{\pi}{16} H_m + \frac{\pi}{4} \log 2 \right\} = \frac{G}{2}. \quad (58)$$

Proof. Write

$$t_j = \frac{16^{j-1}}{j^2 \binom{2j}{j}^2}, \quad F(x) = \sum_{j \geq 1} t_j x^j = \frac{x}{4} {}_3F_2 \left(\begin{matrix} 1, 1, 1 \\ 3/2, 3/2 \end{matrix}; x \right).$$

Euler's integral for one numerator–denominator pair, followed by

$${}_2F_1(1, 1; 3/2; y) = \frac{\arcsin(\sqrt{y})}{\sqrt{y(1-y)}}$$

and the substitution $t = \sin^2 \theta$, gives, for $0 < x < 1$,

$${}_3F_2 \left(\begin{matrix} 1, 1, 1 \\ 3/2, 3/2 \end{matrix}; x \right) = \frac{1}{\sqrt{x}} \int_0^{\pi/2} \frac{\arcsin(\sqrt{x} \sin \theta)}{\sqrt{1 - x \sin^2 \theta}} d\theta. \quad (59)$$

Let $K(x) = \int_0^{\pi/2} (1 - x \sin^2 \theta)^{-1/2} d\theta$. Subtracting $\pi/2$ in the numerator of (59) decomposes its right side as

$$\frac{1}{\sqrt{x}} \left\{ \frac{\pi}{2} K(x) + J(x) \right\}, \quad J(x) = \int_0^{\pi/2} \frac{\arcsin(\sqrt{x} \sin \theta) - \pi/2}{\sqrt{1 - x \sin^2 \theta}} d\theta.$$

The standard endpoint estimate is $K(x) = \log 4 - \frac{1}{2} \log(1-x) + o(1)$. Dominated convergence applies to J : if $a = \arccos(\sqrt{x} \sin \theta)$, the absolute value of its integrand is $a/\sin a \leq \pi/2$. Therefore

$$\lim_{x \uparrow 1} J(x) = - \int_0^{\pi/2} \frac{u}{\sin u} du = 2 \int_0^{\pi/4} \log(\tan v) dv = -2G.$$

It follows that

$$\lim_{x \uparrow 1} \left\{ F(x) + \frac{\pi}{16} \log(1-x) \right\} = \frac{\pi}{4} \log 2 - \frac{G}{2}. \quad (60)$$

Stirling's formula gives $t_j = \pi/(16j) + O(j^{-2})$. Hence the series obtained by subtracting $\pi/(16j)$ is absolutely convergent, and Abel's theorem turns (60) into (56). Rearrangement gives (57).

Finally, the even recurrence in (45), or directly (46), gives

$$\frac{T_{2m}}{\binom{2m}{m}^2} = - \sum_{j=1}^m t_j.$$

Since $H_m - \log m \rightarrow \gamma$, the partial-sum form of (56) proves (58). $\square$

The endpoint $c = 4$ is part of a full transport family. It also explains why ordinary binomial acceleration alone cannot turn this companion into an irrationality proof.

Proposition 6.14 (binomial transport and its optimal acceleration). *For a sequence U , set*

$$U_n^{[c]} = \sum_{k=0}^n \binom{n}{k} (-c)^{n-k} U_k.$$

If $U = A_E$, respectively $U = B_E$, then

$$(n+1)^2 U_{n+1}^{[c]} - (4-c)(3n^2 + 3n + 1) U_n^{[c]} + (32 - 24c + 3c^2) n^2 U_{n-1}^{[c]} + c(4-c)(8-c)n(n-1) U_{n-2}^{[c]} = \begin{cases} 0, & U = A_E, \\ (-c)^n, & U = B_E. \end{cases} \quad (61)$$

Its characteristic polynomial factors as

$$(X - (8-c))(X - (4-c))(X + c). \quad (62)$$

For $c = 0, 1, 2, 3$,

$$\lim_{n \rightarrow \infty} \frac{B_n^{[c]}}{A_n^{[c]}} = \frac{G}{2}, \quad \limsup_{n \rightarrow \infty} \left| \frac{G}{2} - \frac{B_n^{[c]}}{A_n^{[c]}} \right|^{1/n} = \frac{\max(c, 4-c)}{8-c}. \quad (63)$$

Thus $c = 2$ is the unique optimum on $[0, 4)$, improving the convergence factor from $1/2$ to $1/3$. Nevertheless no integral transform in this family produces a linear form tending to zero: the numerator $A_n^{[c]} G/2 - B_n^{[c]}$ has exponential scale $\max(c, 4-c) \geq 2$ for $c = 0, 1, 2, 3$, while $c = 4$ has scale 4 and the logarithmic collision displayed in (58).

Proof. Ordinary binomial convolution gives

$$U^{[c]}(z) = \frac{1}{1+cz} U\left(\frac{z}{1+cz}\right).$$

Substitute this identity into (47); the chain rule gives (61). The homogeneous leading part factors as (62).

For completeness, the nonvanishing of the relevant asymptotic components can be read without a connection calculation. The bounds (37) and $A_E(n) \leq 8^n$ give $A_E(n)^{1/n} \rightarrow 8$. The positive error formula (40), together with the recurrence, gives

$$\lim_{n \rightarrow \infty} |A_E(n)G/2 - B_E(n)|^{1/n} = 4.$$

Under the generating-function substitution, these two singular scales move to $8-c$ and $4-c$. The latter singularity cannot disappear because the prefactor is regular and nonzero at its preimage. In addition, the right side of (61) forces the simple characteristic root $-c$ in the transformed linear form. Poincaré asymptotics therefore give (63). Finally,

$$\frac{\max(c, 4-c)}{8-c} = \begin{cases} (4-c)/(8-c), & 0 \leq c \leq 2, \\ c/(8-c), & 2 \leq c < 4, \end{cases}$$

which decreases to $1/3$ and then increases. The asserted raw scales and the endpoint case follow at once. $\square$

Theorem 6.15 (half-digit twisted Lucas law). *Let p be an odd prime and let $0 \leq a, r < p$ with $2a + 1 < p$. In the localisation $\mathbb{Z}_{(p)}$,*

$$p^2 B_E(ap + r) \equiv \chi_{-4}(p) B_E(a) A_E(r) \pmod{p}. \quad (64)$$

Proof. Write $k = bp + s$, $0 \leq s < p$, and extend the shell by zero outside $0 \leq k \leq n$. Lucas' theorem, applied also to the two central binomial coefficients, gives

$$S_E(ap + r, bp + s) \equiv S_E(a, b) S_E(r, s) \pmod{p}.$$

Since $2a + 1 < p$, every harmonic argument in a contributing cell is strictly below p^2 . Thus multiplying a weight- q letter by p^q makes it p -integral.

If $S_E(r, s) \not\equiv 0 \pmod{p}$, then Kummer's carry criterion gives

$$s \leq r, \quad 2s < p, \quad 2(r - s) < p.$$

Consequently

$$\left\lfloor \frac{k}{p} \right\rfloor = b, \quad \left\lfloor \frac{2k}{p} \right\rfloor = 2b, \quad \left\lfloor \frac{2(n - k)}{p} \right\rfloor = 2(a - b).$$

Every monomial in the weight in (20) has total weight 2 and character signature χ_{-4} . Lemma 5.2 therefore gives

$$p^2 W(ap + r, bp + s) \equiv \chi_{-4}(p) W(a, b) \pmod{p}.$$

If $S_E(r, s) \equiv 0$, the shell congruence and p -integrality of $p^2 W$ make both sides of the corresponding summand congruence zero. Summing over the full digit block separates the b - and s -sums and proves (64). $\square$

Lemma 6.16 (the last digit). *For every odd prime p ,*

$$A_E(p - 1) \equiv (-1)^{(p-1)/2} = \chi_{-4}(p) \pmod{p}. \quad (65)$$

Proof. Put $h = (p - 1)/2$. In the sum defining $A_E(p - 1)$, the factor $\binom{2k}{k}$ vanishes modulo p when $k > h$, while $\binom{2p-2-2k}{p-1-k}$ vanishes when $k < h$. Only $k = h$ remains. Since $\binom{p-1}{h} \equiv (-1)^h \pmod{p}$, its shell value is $(-1)^{3h} = (-1)^h$. The last equality in (65) is Euler's criterion for the quadratic character modulo 4. $\square$

Theorem 6.17 (unit-digit full Lucas law). *Let p be an odd prime such that*

$$p \nmid A_E(j) \quad (0 \leq j < p). \quad (66)$$

Then for every $0 \leq a, r < p$ the unrestricted congruence

$$p^2 B_E(ap + r) \equiv \chi_{-4}(p) B_E(a) A_E(r) \pmod{p} \quad (67)$$

holds in $\mathbb{Z}_{(p)}$.

Proof. Write $N = ap + r$. Summing the Casoratian increments in the proof of Corollary 6.7, but stopping at N , gives the rational identity

$$\frac{B_E(N)}{A_E(N)} = \sum_{m=1}^N \frac{32^{m-1}}{m^2 A_E(m) A_E(m-1)}. \quad (68)$$

Summing the shell factorisation used in the proof of Theorem 6.15 gives $A_E(cp + s) \equiv A_E(c) A_E(s) \pmod{p}$. Under (66), every $A_E(m)$ with $0 \leq m < p^2$ is therefore a p -unit. After multiplying

(68) by p^2 , all summands with $p \nmid m$ vanish modulo p . Writing the remaining indices as $m = jp$, Fermat's theorem and Lucas descent give

$$\begin{aligned} \frac{p^2 B_E(N)}{A_E(N)} &\equiv \sum_{j=1}^a \frac{32^{j-1}}{j^2 A_E(j) A_E(j-1) A_E(p-1)} \\ &= \frac{B_E(a)}{A_E(a) A_E(p-1)} \pmod{p}. \end{aligned}$$

Finally $A_E(N) \equiv A_E(a) A_E(r)$, and $A_E(p-1)^{-1} \equiv \chi_{-4}(p)$ by Lemma 6.16. This proves (67). $\square$

Corollary 6.18 (optimality of the denominator exponent). *For every prime $p \geq 5$,*

$$v_p(B_E(p)) = -2. \quad (69)$$

Consequently the exponent 2 in Theorem 6.9 cannot be lowered in any uniform bound $L_n^e B_E(n) \in \mathbb{Z}$.

Proof. Take $a = 1$ and $r = 0$ in Theorem 6.15; its half-digit condition is $3 < p$. Since $B_E(1) = A_E(0) = 1$, it gives $p^2 B_E(p) \equiv \chi_{-4}(p) \not\equiv 0 \pmod{p}$. Theorem 6.9 already gives $v_p(B_E(p)) \geq -2$, so equality follows. $\square$

Remark 6.19 (the full-digit carry problem). Exact recurrence computation gives (64) for every $0 \leq a, r < p$, without the half-digit restriction, for all odd primes $p \leq 31$. The unrestricted statement does not follow termwise from Theorem 5.3: when $2k$ crosses p^2 , individual carry cells develop p -adic poles. Their sum nevertheless cancels in every test. Proving this second-digit carry cancellation would remove the residual zero-digit condition not covered by Theorem 6.17 and yield a full companion Lucas law.

Remark 6.20 (irrationality boundary). The formula is a new finite harmonic representation of the normalized second solution, but not an irrationality proof for G . The characteristic roots of (2) are 8 and 4; the complementary linear form grows on the 4^n scale instead of tending to zero, and clearing harmonic denominators only worsens the arithmetic estimate. This is consistent with the obstruction in earlier Apéry-like constructions for Catalan's constant [16]. Theorem 6.9 improves the clearing factor from the elementary $L_{2n}^2 = \exp((4 + o(1))n)$ bound to $L_n^2 = \exp((2 + o(1))n)$, but the resulting linear form still has size on the $4^n L_n^2$ scale. Thus the new sharp denominator theorem does not by itself imply irrationality. Proposition 6.14 sharpens this statement: even the optimal fixed integral binomial acceleration has a raw small-form scale 2^n , before any denominator is cleared.

7 Algorithm and further directions

The results suggest the following reproducible search-and-proof pipeline.

1. Fit a candidate weight in a graded alphabet of ordinary and residue-coloured harmonic letters. Separate character signatures from the beginning.
2. Convert every fitted monomial to its connected lift by Theorem 2.3. At weight two use Corollary 2.4; compact several blocks by matching logarithmic derivatives when possible.
3. Split indices only by the least common period needed for cyclotomic letters. Theorem 3.1 then produces proper hypergeometric cells.
4. Telescope one-parameter polarizations such as (9), differentiate their recurrences, and eliminate lower jets. Verify the rational certificates before using any initial values.
5. Test Theorem 5.3 termwise. If digit compatibility fails, the obstruction identifies exactly which wide argument or carry correction must be repaired.

Two directions look particularly worthwhile.

Higher characters. The residue-indicator construction treats non-real characters without fractional hypergeometric exponents. It should make weight-two companions for cubic characters as accessible as the χ_{-4} example, at the cost of a larger finite residue split.

Minimal operators from connected lifts. The universal lift proves existence but can produce a large Ore operator. Connected lifts contain much more cancellation than a generic parameter derivative. A systematic criterion for when those cancellations force the original Apéry operator, rather than a left multiple, would turn many experimentally minimal harmonic formulas into short proofs.

Multi-digit divided-power jets. The single-digit bound in Theorem 5.3 is honest. A natural replacement for iteration is to package the parameter Taylor series in divided powers, where denominators and successive Frobenius descents can be tracked simultaneously. This may connect the elementary harmonic-jet picture with unit-root and crystal formulations of Apéry congruences.

8 Reproducibility

The accompanying script `work/harmonic_jets/check_lifts.py` uses only Python’s integer and rational arithmetic. It checks the logarithmic-derivative decomposition in Proposition 6.2 cell by cell for $n \leq 19$ and audits (20) and (2) through $n = 30$. The script `work/harmonic_jets/verify_catalan_scalar.py` proves the scalar input recurrences symbolically from (30). Finally `work/harmonic_jets/prove_catalan_ore.py` reconstructs the common factor, verifies the exact right divisions, and performs the finite propagation checks in Proposition 6.5; it runs in the open-source `PassageMath/ore.algebra` environment. The Maxima file `work/harmonic_jets/explore_e.mac` records the parity-split setup, with two-dimensional display disabled. The exact audit `work/harmonic_jets/check_catalan_lucas.py` checks the unrestricted congruence in Remark 6.19 for every odd prime through 31 and reports which of them satisfy the proved unit-digit hypothesis. The script `work/harmonic_jets/check_catalan_endpoint_transform.py` checks (61) for $0 \leq c \leq 8$, (45), the square expansion (46), and the termwise divisibility $L_n R_{n,j} \in \mathbb{Z}$ through $n = 300$. The SymPy script `work/harmonic_jets/verify_catalan_transport_symbolic.py` verifies the generating-function chain rule and characteristic factorisation for symbolic c . The independent script `work/harmonic_jets/verify_general_endpoint.py` verifies (51) for symbolic b , audits (52) through $n = 120$ for several positive and negative parameters divisible by 4, and derives both necessity witnesses in (55) symbolically. The older independent recurrence audit `work/harmonic_jets/check_catalan_denominators.py` verifies Theorem 6.9 directly through $n = 500$. A detailed Lean handoff is provided in `work/harmonic_jets/LEAN_CATALAN_HANDOFF.md`.

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